Amari Jaconelli

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Research Interests

My current work is primarily concerned with: random matrix models, integrable systems, operator determinants, point processes, and Riemann-Hilbert techniques. I also have a physics background in complex and heterogeneous systems.

Education

05/2025 – 07/2025
Angers, France

LAREMA | Research Placement
Université d'Angers

- Research Project: Connecting a class of Fredholm Determinant relating to NLS to a Painlevé IV Vector System
- Host: Dr. Mattia Cafasso

01/2024 – present
Bristol, United Kingdom

School of Mathematics | Doctor of Philosphy
University of Bristol

- Proposed Thesis Title: 'From Non-Hermitian to Hermitian Random Matrices'
- Supervisor: Dr. Thomas Bothner

09/2019 – 07/2023
Coventry, United Kingdom

School of Physics | Masters of Physics - First Class
University of Warwick

- Research Project: 'Active Smectic Liquid Crystals through Magnetic Analogy'
- Supervisor: Dr. Gareth Alexander

06/2022 – 09/2022
Coventry, United Kingdom

HetSys | Research Intern
University of Warwick

- Research Project: 'BFGS Minimisation using a Gaussian Regression Model for the Hessian'
- Supervisors: Dr. Albert Bartok-Partay, Prof. James Kermode

Teaching Experience

01/2025 – 04/2025
University of Bristol - School of Mathematics

Support Session Leader
Quantum Information Theory

Third Year Module taught by the Bristol School of Mathematics.

01/2025 – 04/25
University of Bristol - School of Mathematics

Support Session Leader
Fields, Forms, and Flows

Third Year Module taught by the Bristol School of Mathematics.

11/2024 – 07/2025
University of Bristol - School of Mathematics

Maths Tutor
We Solve Problems

WSP is a charity which endeavours to improve maths education for secondary school children across the UK by providing additional weekend problem-solving classes out of local universities. My role is to teach and supervise at these sessions.

09/2024 – 12/2025
University of Bristol - School of Mathematics

Teaching Assisstant
Mathematical Programming

Second Year Module taught by the Bristol School of Mathematics.

11/2023 – 09/2024

Online Mathematics & Physics Tutor
Redford Education

This is an online tutoring service which focuses on GCSE, A Levels, and private entrance exams.

Attended Conferences and Schools

12/2026
XXII Brunel–Bielefeld Workshop on RMT
Brunel University of London
London, United Kingdom - Participant

09/2026
Workshop on Random Matrices and Integrability
University of Oxford
Oxford, United Kingdom - Participant and Speaker

08/2026
Orthogonal Polynomials, Special Functions, and Applications - 18
Doshisha University
Kyoto - Participant and Speaker

06/2026
Interacting Particle Systems Related to Random Matrices
Univeristy of Manchester
Manchester - Invited Participant

06/2026
Integrable systems: Regularity, Non-Commutativity, and Random Matrix Theory Workshop
International Centre for Mathematical Sciences (ICMS)
Edinburgh, United Kingdom - Invited Participant and Speaker

05/2026
Configurations de points : de la physique statistique à la théorie du potentiel
Centre International de Rencontres Mathématiques (CIRM)
Marseille, France - Participant and Speaker

08/2025
5th ZiF Summer School: Randomness in Physics and Mathematics
Universität Bielefeld
Bielefeld, Germany - Participiant and Speaker

07/2025
InterPlay between Integrable Probability and Interacting Particle Systems
Université d'Angers
Angers, France - Participant

06/2025
'Integrability, Random Matrices, and All That' - PIICQ Network Conference
ENS de Lyon
Lyon, France - Participant

05/2025
L'esprit des Cartes: une Conférence en l'Honneur d'Emmanuel Guitter
Université Paris-Saclay
Saclay, France - Participant

05/2025
EWM Nordic Conference: Women in Mathematics
Kungliga Tekniska Högskolan
Stockholm, Sweden - Participant

12/2024
XX Brunel-Bielefeld Workshop on Random Matrix Theory and Applications
Brunel University of London
London, United Kingdom - Participant

10/2024
Heilbronn Institue of Mathematical Research: Averages of Deratives of Characteristic Polynomials of Random Matrices; Focused Research Workshop
University of Bristol
Bristol, United Kingdom – Participant

06/2024
Random Matrix Theory Summer School
University of Michigan
Ann Arbor, United States of America – Participant

Publications and Preprints

11/2025

On Fredholm Pfaffians and Riemann-Hilbert problems

https://arxiv.org/abs/2511.23362

Currently out on preprint. To be published in an upcoming special volume on the occasion of Percy Deift's 80th birthday in the KIAS Springer Series in Mathematics.

Talks

10/2026

ISRMT Seminar

University of Michigan

Title: 'On Fredholm Pfaffians and Riemann-Hilbert problems'

03/2026

PIICQ Seminar

Institut de Mathématiques de Jussieu-Paris Rive Gauche

Title: 'On Fredholm Pfaffians and Riemann-Hilbert problems'